Lecture 3 - Analysis of Sets

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Theorem 1. *The set* $\mathbb{N} \times \mathbb{N} = \{(n, m) \mid n, m \in \mathbb{N}\}$ *is denumerable.*

Proof. Informally, we can show this with a diagonalization proof:

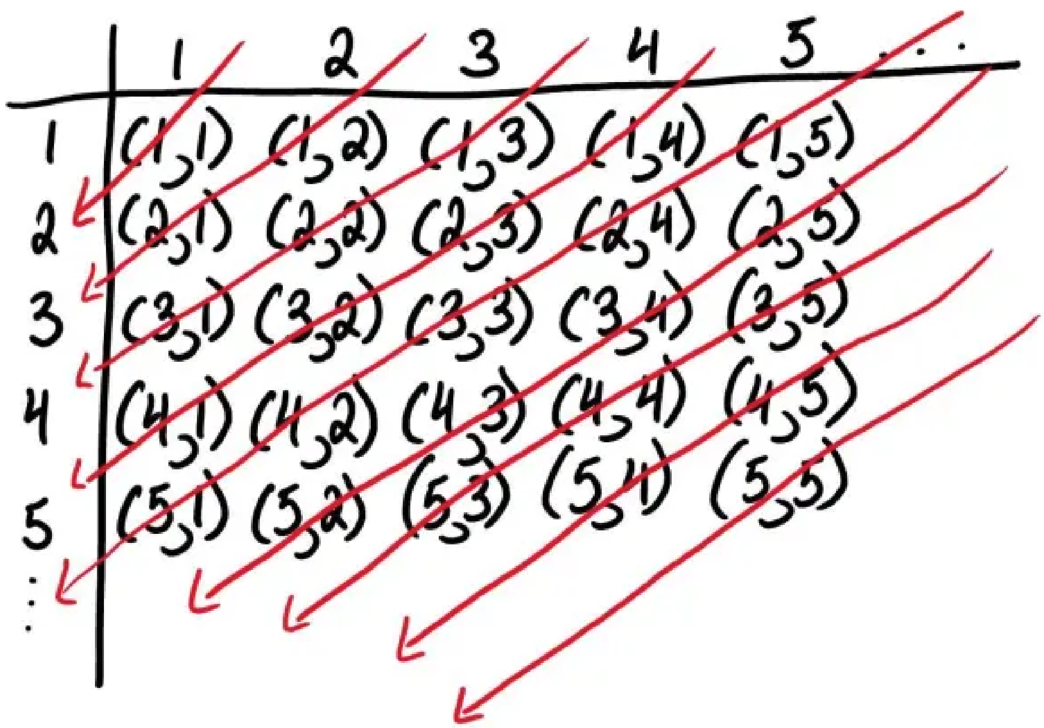

Figure 1: Diagonalization argument for countability of $\mathbb{N} \times \mathbb{N}$

Here, each entry in the table corresponds to an element of the set $\mathbb{N} \times \mathbb{N}$. Since we are able to draw a line (the red arrows), which crosses through each element in the set exactly once, this informal proof shows us that there exists a bijective function between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$.

□

Corollary 1.1. *The set* $\mathbb{Q}$ *is denumerable.*

Proof. The proof for the denumerability of $\mathbb{Q}$ can be demonstrated with a diagonalization argument much in the same way as the proof for $\mathbb{N} \times \mathbb{N}$.

□

Theorem 2. *Russel's Paradox: There does not exist a set of all sets.*

Proof. Assume, for the purposes of contradiction, that there exists a set of all sets. Then, in such a set, there must exist a set A which contains all sets which do not contain themselves. Now, if A does not contain itself, then A is a set which does not contain itself and thus must belong to A. But then, if A is not contained in A, then A is a set which does not contain itself and therefore should be in A. This is necessarily a contradiction and thus no such set can exist □

- The implication of Russell's paradox is that there must exist certain requirements on what can and cannot be a set.
- For two sets A and B, B is a *subset* of A if $b \in B \implies b \in A$.
  - Alternatively, we can call A a *superset* of B.
- For a set A, the power set of A is the set of all subsets of A.
  - Denoted $\wp(A)$.

Theorem 3. *Cantor's Theorem: For any set A, there exists not surjective function f such that $f : A \to \wp(A)$.*

Proof. We begin by separating our proof into 2 cases depending on the cardinality of A.

If A is finite

Between two finite sets A and B, there can exist no surjection from A to B if $\#A < \#B$. If A is finite, then we know there exists a bijection between A and $\mathbb{N}_n$ for some $n \in \mathbb{N}$, making the cardinality of A n. The cardinality of the power set of A is therefore given by 2^n (since $\forall a \in A$, we have a binary choice between whether to include it or not in a given subset of A).

Using this, we know that there can exist no surjection from A to $\wp(A)$ if $\#A < \#\wp(A)$. Since $\#A = n$ and $\#\wp(A) = 2^n$, we need to prove $P(n) : n < 2^n$ is true $\forall n \in \mathbb{N}$, which we do by induction.

(1) Prove the base case (Show $P(1)$ is true).

LHS:	RHS:
n	2^n
$= 1$	$= 2^1$
$= 1$	$= 2$

Therefore, since $LHS < RHS$, $P(1)$ is true.

(2) Formulate the induction hypothesis (State $P(k)$).

$$P(k) : k < 2^k$$

(3) Prove $P(k) \implies P(k + 1)$

$$\begin{aligned} &k+1 \\ &< 2^k + 1 \\ &< 2^{k+1} \end{aligned}$$

Therefore, since $P(1)$ is true and $P(k) \implies P(k+1)$, by the principle of mathematical induction, $P(n)$ is true $\forall n \in \mathbb{N}$.

Thus, our proof for finite A is complete, since $\#A = 2 < 2^n = \#\wp(A)$ implies there can exist not surjective function from A to $\wp(A)$.

If A is infinite

$\square$

Corollary 3.1. *If A is denumerable then $\wp(A)$ is uncountable.*